

Series and infinite products – 5

Double and multiple series

Homework

12 сентября 2026 года

Theoretical minimum. For non-negative terms a double sum may be computed in any order and equals the supremum over finite partial sums; in the general case one first checks absolute summability, and without it the two iterated sums may differ. The Cauchy product of two series is legitimate as soon as one of them converges absolutely (Mertens); for two conditionally convergent series it may fail. Rearranging $\mathbb{N}^2$ by the level sets of mn or $m+n$ produces arithmetic weights such as the divisor function. A factor depending on $n_1 + \dots + n_k$ is made separable by $\frac{1}{n_1 + \dots + n_k + 1} = \int_0^1 x^{n_1 + \dots + n_k} dx$, after which the multiple sum becomes an integral of a power of a single sum. Splitting $\mathbb{N}^2$ into the three parts $m > n$, $m < n$, $m = n$ gives the relations between multiple zeta values.

1. Prove Mertens' theorem, and show that for $a_n = b_n = \frac{(-1)^n}{\sqrt{n+1}}$ the Cauchy product diverges.
2. Prove that $\sum_{m,n \geq 1} \frac{1}{mn(m+n)} = 2\zeta(3)$.
3. Prove Euler's identity $\zeta(2, 1) = \zeta(3)$, that is $\sum_{m > n \geq 1} \frac{1}{m^2 n} = \zeta(3)$, and deduce $\zeta(2)^2 = 2\zeta(2, 2) + \zeta(4)$.
4. For which s does $\sum_{(m,n) \neq (0,0)} \frac{1}{(m^2 + n^2)^s}$ converge? Prove the answer by counting lattice points in annuli, and state the analogue in $\mathbb{Z}^d$.
5. Prove that $\sum_{n \geq 1} \frac{d(n)}{n^s} = \zeta(s)^2$ for $s > 1$, where $d(n)$ is the number of divisors, and that $\sum_{n \geq 1} \frac{x^n}{1-x^n} = \sum_{n \geq 1} d(n)x^n$ for $|x| < 1$.

6. Prove that for every fixed $m \geq 1$ $\sum_{n \geq 1, n \neq m} \frac{1}{m^2 - n^2} = \frac{3}{4m^2}$, and use it to show that the double family $\left(\frac{1}{m^2 - n^2}\right)_{m \neq n}$ is not absolutely summable while both iterated sums exist.
7. Let a_{mn} be the matrix with $a_{nn} = 1$, $a_{n+1,n} = -1$ and zeros elsewhere. Compute both iterated sums and explain, in the language of Theorem 1, exactly which hypothesis fails.
8. Prove that for $a_{mn} \geq 0$ the sum over $\mathbb{N}^2$ does not depend on the exhaustion of $\mathbb{N}^2$ by finite sets, and give an example of a family for which two different exhaustions produce two different limits.
9. Compute $\sum_{m,n \geq 1} \frac{(-1)^{m+n}}{m+n}$ as an iterated sum in both orders, and decide whether the family is summable.
10. (OMOUS 2024) Compute the limit $\lim_{n \rightarrow \infty} \frac{\sum_{i,j,k=1}^n \frac{1}{i+j+k}}{n^2 \ln n}$.